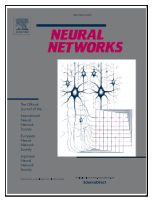
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Full Length Article



DiffuMSMO: A multi-modal diffusion model for extractive multi-modal summarization with multi-modal output

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ABSTRACT

Extractive multi-modal summarization with multi-modal output (MSMO) aims to form a multi-modal summary by extracting sentences and images from the source multi-modal document. Existing works mostly formulate it as two sequence labeling problems by making individual sentence label prediction and image label prediction. This paper proposes a novel extractive multi-modal summarization method named DiffuMSMO, by directly generating the desired summary image representations with an ImageText2Image diffusion model and generating summary sentence representations with an ImageText2Text diffusion model, and then extracting images and sentences based on image representation matching and sentence representation matching. To bridge the gap between sentences and news images, DiffuMSMO fine-tunes a news-domain-specific CoCa model on a news image captioning dataset. A multi-modal Transformer encoder is optimized with a contrastive image matching loss, a contrastive sentence matching loss, and a contrastive sentence-image alignment loss for contextualized image and sentence encoding. Experiments on the MSMO dataset demonstrate that DiffuMSMO achieves competitive performance. It outperforms nearly all extractive and abstractive summarization baselines across all ROUGE and IP metrics. When compared with the strongest baseline, DiffuMSMO achieves statistically significant improvements on R-1 and R-L, while the baseline yields better results on R-2 and IP metrics. Ablation study shows effectiveness of each part of the model. The results demonstrate the great potential of adapting diffusion models for extractive MSMO.

1. Introduction

Multi-modal summarization with multi-modal output (MSMO) aims to create a multi-modal summary with a summary text and several summary images for a multi-modal document (Zhu et al., 2018). As with pure text summarization, the MSMO methods can also be divided into two types. Abstractive MSMO generates text abstractive summary text as the text summary and suffers from problems of generating ungrammatical or even nonfactual contents (Chen & Zhuge, 2019; Chen & Zhuge, 2018). Extractive MSMO extracts significant sentences as the text summary and is grammatically accurate and faithful. For both types of MSMO methods, image summaries are extractive by extracting significant images which are aligned with the summary sentences to form multi-modal summaries.

This work focuses on extractive MSMO. Extractive MSMO can be formulated as a sentence labeling sub-task and an image labeling sub-task, which predict a 0/1 label for each sentence and each image indicating whether or not the sentence or the image should be included in the multi-modal summary (Chen & Zhuge, 2019). In contrast, abstractive MSMO formulates text summary generation as a text generation subtask (Zhu et al., 2018; Zhang et al., 2024). However, abstractive MSMO is not fully abstractive, because it formulates image summary generation as an image labeling sub-task. It is difficult to synthesize high-quality abstractive summary images from long and multi-modal documents. The synthesized low-quality images can not be directly used as summary images. It has been shown that text generation information can be used to guide sentence extraction for extractive text summarization through diffusion models (Zhang et al., 2023a). However, how to use image generation models for multi-modal summarization has not been explored. We argue that the synthesized images contain useful summary information and can be used as guidance for image selection and sentence selection in extractive multi-modal summarization.

Diffusion models are recently developed and show efficiency in text2image applications (Saharia et al., 2022b) and text2text applications (Li et al., 2022). Zhang et al. (2023a) propose the diffusum model by enhancing extractive summarization with diffusion models. MSMO is different from text summarization in that the input is a multi-modal document, and the output is a text summary accompanied with images.

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Most diffusion models input single-modal data and output single-modal products. For example, diffusion-based text2image models input text data and output images (Batzolis et al., 2021; Batzolis et al., 2021), diffusion-based text2video models input text data and output videos (Khachatryan et al., 2023), image2image models input images and output images (Saharia et al., 2022a), and image2video models input image data and output videos (Xing et al., 2025). For diffusion-based text summarization, such as diffusion-lm (Li et al., 2022), diffuseq (Gong et al., 2022), seqdiffuseq (Yuan et al., 2022), and diffusum (Zhang et al., 2023a), the input and the output are both text. The conditional diffusion model first uses text as the condition in the diffusion process for the text2image task (Zhang et al., 2023b). DiffuSeq proposes a diffusion model designed for seq2seq text generation tasks by adding partial noise during the forward process and conditional denoising during the reverse process. DiffuSum follows DiffuSeq to use source text as conditions in the denoising process to generate desired summary sentence representations and extracting sentences based on sentence representation matching. DiffuSum is the diffusion-based model for extractive text summarization, and its inputs and outputs are both text modalities. This paper follows DiffuSum by extending the diffusion model to the multi-modal settings of MSMO. There are two problems. One is to deal with multi-modality of input in the noising process and denoising process, and the other is to output multi-modal summaries of MSMO based on denoised representations.

Hence, this paper proposes a two-line multi-modal diffusion model for extractive MSMO (DiffuMSMO). DiffuMSMO generates summary sentence representations and summary image representations with two lines of diffusion models conditioned on source sentences and source images, and then extract summary sentences and summary images by matching with the generated representations. The first line of DiffuMSMO is the ImageText2Text diffusion line which generates the continuous summary sentence representations, and extracts summary sentences based on sentence representation matching. The second line of DiffuMSMO is the ImageText2Image diffusion line which generates continuous summary image representations, and extracts summary images based on image representation matching. Both lines of diffusion models add partial noise during the forward process, and use source text and images as conditions in the denoising process to generate summary sentence representations and summary image representations. Moreover, DiffuMSMO incorporates a contrastive image-sentence alignment module, an image matching module and a sentence matching module with multi-class contrastive loss (Khosla et al., 2020) for multi-modal representations. DiffuMSMO jointly optimizes the ImageText2Text diffusion line and the ImageText2Image diffusion line, and extracts summary sentences and summary images by representation matching without any rounding step. Experiments are carried out in the MSMO dataset, and the results demonstrate that DiffuMSMO achieves a comparable or even better performance than state-of-the-art systems such as pre-trained models and diffusion-based text summarization models. Ablation study also shows the effectiveness of either line of DiffuMSMO and the effectiveness of the sentence-image alignment module for multi-modal summarization.

The contributions of the work can be summarized as follows:

(1) This paper proposes the two-line multi-modal diffusion-based extractive MSMO framework DiffuMSMO by extending diffusion models to multi-modal settings of MSMO. DiffuMSMO consists of an ImageText2Text diffusion model and an ImageText2Image diffusion model, using text and images as inputs to generate continuous summary image representations and summary sentence representations, and extract sentences and images based on representation matching. To our best knowledge, we are the first to apply diffusion models to the MSMO task.
(2) This paper introduces a contrastive sentence-image alignment module, a contrastive image matching module, and a sentence matching module in DiffuMSMO for multi-modal representation generation.
(3) This paper carries out extensive experiments on the MSMO dataset and show that DiffuMSMO outperforms baseline extractive and abstractive summarization models according to ROUGE scores and IP scores. Ablations demonstrate effects of each part of DiffuMSMO.

2. Related work

2.1. Multi-modal summarization

Multi-modal summarization aims to distill concise and precise syntheses from multi-modal data sources such as textual, visual, and audio data, etc (Zhu et al., 2018). Some works (Li et al., 2017; Li et al., 2018; Chen & Zhuge, 2018; Zhuge, 2020; Zhu et al., 2023; Wu et al., 2020; Chen & Zhuge, 2014) augment the depiction of a solitary modality with supplementary modal information. For example, Chen and Zhuge (2018) augment the semantic richness of the input sentence by employing attentional mechanisms to harness textual and visual information in images and captions.

Recently, there has been a growing interest in the domain of multi-modal summarization with multi-modal output (MSMO), which integrates textual and visual inputs to produce a pictorial summary. The large-scale corpus MSMO for the task, and a multi-modal attention-based framework is proposed to generate textual summary and to select the most relevant images (Zhu et al., 2018). They also propose a novel evaluation metric to integrate visual and textual content during both the training stage and the assessment stage. The MSMO task is proposed to solve in a multi-task training manner (Mukherjee et al., 2022; Zhang et al., 2022a), and a graph network and a hierarchical fusion framework are employed to learn the intra-modal and inter-modal correlations in the multi-modal data (Zhang et al., 2022b). An entity-guided multi-modal summarization model is proposed to integrate the fine-grained entity knowledge (Zhang et al., 2024), which is built on BART (Lewis et al., 2020) by utilizing dual multi-modal encoders with shared weights to process text-image and entity-image information. Other works construct heterogeneous graphs that use sentences, words and images as nodes from multi-modal documents, and employ GAT (Xie et al., 2023) or Graphormer (Jiang & Chen, 2025) to create extractive multi-modal summaries.

2.2. Diffusion models on text and image

Diffusion models are first proposed in the continuous domain (Ho et al., 2020; Ramesh et al., 2022) and have recently gained attention in image generation (Batzolis et al., 2021), video generation (Xing et al., 2025), audio generation (Schneider, 2023), and text generation (He et al., 2023).

For text generation, Diffusion-lm (Li et al., 2022) adapts continuous diffusion models for text by adding an embedding step and a rounding step, DiffuSeq (Gong et al., 2022) designs a diffusion model for seq2seq text generation tasks by adding partial noise during the forward process and conditional denoising during the reverse process. For text summarization, DiffuSum (Zhang et al., 2023a) explores diffusion models for extractive summarization by generating summary representations and selecting sentences based on the matching of sentence representations. Additionally, a contrastive sentence encoding module is introduced in DiffuSum to learn sentence representations, further enhancing the performance of DiffuSum. TermDiffuSum further incorporates legal terminology into the diffusion model by allocating higher attention weight to sentences containing a higher concentration of legal terms for extractive summarization of legal documents. For image captioning, the typical image2text task, SCD-Net (Luo et al., 2023) is a diffusion model based paradigm tailored for image captioning. In SCD-Net, multiple diffusion Transformer structures are stacked to progressively strengthen the output sentence with better vision-language alignment, and a self-critical sequence training strategy is designed to stabilize the diffusion process.

For image generation, ControlNet (Zhang et al., 2023b) adds spatial conditioning controls to pretrained text2image diffusion models by locking the production-ready large diffusion models and learning a diverse set of conditional controls. Saharia et al. (2022a) develops a unified framework for image2image translation based on conditional diffusion models and evaluates the framework on image2image translation tasks. For video generation, Khachatryan et al. (2023) introduce the task of zero-shot text2video generation and propose a low-cost approach by leveraging the power of existing text2image synthesis methods. Xing et al. (2025) propose the Multi-Modal Large Language Model for image2video to predict future video states based on initial frames and text instructions.

The above diffusion-based models input single-modal data such as text and images, and output text, images or videos. In particular, the diffusion-based text summarization model DiffuSum inputs text and output text. Our diffusion-based MSMO model differs from them as follows: 1) The input of the model is multi-modal data, i.e., text and images, and the output of the model are also text and images, so our model is divided into a ImageText2Text part and a ImageText2Image part. 2) We incorporate a text-image alignment module to improve multi-modal representations for multi-modal summarization.

3. Proposed model

This section provides a detailed examination of DiffuMSMO, as illustrated in Fig. 1. DiffuMSMO mainly consists of a multi-modal Transformer-based encoder and a two-line diffusion module. Firstly, the initial embeddings of sentences and images are obtained using the CoCa model tuned in the news domain (Section 3.2). Secondly, the sentence and image embeddings are input to the multi-modal Transformer encoder to interact with each other for encoding, and compute image matching loss, sentence matching loss and sentence-image alignment loss for training (Section 3.3). Thirdly, the sentence and image encodings are passed to the sentence diffusion line and the image diffusion line, and the diffusion losses are obtained (Section 3.4). The whole model is trained by combining the losses. For inference, summary sentences and summary images are randomly initialized and are denoised to create summary sentence representations and summary image representations, and sentences and images are extracted by matching with the denoised representations (Section 3.5). We first define the problem (Section 3.1), and then introduce each part of the model in the following.

3.1. Problem definition

Given a multi-modal document with N sentences and M images (pictures) denoted as $D = \{s_1^{\mathrm{d}}, s_2^{\mathrm{d}}, \ldots, s_N^{\mathrm{d}}, p_1^{\mathrm{d}}, p_2^{\mathrm{d}}, \ldots, p_M^{\mathrm{d}}\}$, extractive MSMO aims to form a multi-modal summary denoted as $S = \{s_1^{\mathrm{s}}, s_2^{\mathrm{s}}, \ldots, s_n^{\mathrm{s}}, p_1^{\mathrm{s}}, p_2^{\mathrm{s}}, \ldots, p_m^{\mathrm{s}}\}$, by directly extracting n ($n \ll N$) sentences and m ($m \ll M$) images from the source multi-modal document. The task can be formulated as two sequence labeling tasks i.e. sentence labeling and image labeling. Each sentence or image is given a 0/1 label where 1 indicates that the sentence or the image will be included in the summary S.

However, in the dataset for the MSMO task, neither ground-truth sentence labels nor image labels are available for training. Human-written abstractive text summaries are provided in the dataset for training and testing, and manually annotated image labels are only available for testing. To get ground-truth sentence labels, it is common to use a greedy algorithm to generate an ORACLE by maximizing the ROUGE-2 score against the gold text summary following Nallapati et al. (2017). To get image labels for training, we fine-tune the CoCa model (Yu et al., 2022) in a news image captioning dataset, and greedily generate an image summary ORACLE by maximizing the TunedCoCa-based sentence-image alignment scores against the gold text summary as follows. Step 1: set A as the set of all summary sentences, and set B as an empty summary image set; Step 2: for each summary sentence $i \in A$, set its aligned image j as $\arg\max(\mathrm{TunedCoCa}(i, j))$; Step 3: set the alignment score of each aligned image j as the number of summary sentences in A that are aligned to j, plus the mean CoCa score of j with all the summary sentences in A; Step 4: select the image j with the highest alignment score, set $B = B \cup \{j\}$, and eliminate from A the summary sentences that are aligned to j; Step 5: if A is not empty go to Step 2, else return B as the ORACLE image summary.

3.2. Fine-tune CoCa for news-domain-specific sentence-image matching

CoCa is an image-text encoder-decoder model jointly pre-trained with contrastive loss and captioning loss, thereby subsuming model capabilities from contrastive approaches like CLIP (Radford et al., 2021) and generative methods. CoCa is pretrained on both web-scale alt-text data and general image classification datasets. To fit CoCa to news-domain-specific image-text matching, we propose to fine-tune CoCa in news image captioning datasets such as NYTimes800K (Tran et al., 2020).

We collect a set of positive image-text pairs from the news captioning dataset for contrastive learning, including both standard-positive and potential-positive samples. Each pair within the standard positive sample set comprises an image and its corresponding ground truth caption. Potential-positive samples are designed to broaden the CoCa model's recognition capabilities, augmenting its proficiency in processing and comprehending sentences derived from actual news articles. For acquiring potential-positive samples, we adopt the advanced pre-trained language model Sentence-Bert (Reimers & Gurevych, 2019) as a teacher network. Concretely, for an image and its human-written caption, Sentence-Bert is used to compute matching scores of the caption and all the sentences, and the sentences are selected as potential-positive samples whose matching scores are greater than η. After obtaining the positive samples, we categorize news captions and sentences lacking correspondence with the image as negative samples, creating a clear definition between matching and mismatching cases. Loss defined in Eqs. (1) and (2) is used to fine-tune CoCa, achieving the knowledge distillation of Sentence-Bert (Xu et al., 2024).

$$\mathcal{L}_{\mathrm{coca}} = -\sum_{j=1}^{n} \omega_j \cdot \log\left(\frac{\exp(\mathrm{logit}_j)}{\sum_{i=1}^{n} \exp(\mathrm{logit}_i)}\right) \tag{1}$$

$$\omega = e^{\frac{\log(\eta)}{\eta - 1}(x-1)}, x \in \mathrm{POS}_j \tag{2}$$

In the equations, POS_j is the matching scores of the positive samples of the image j. The matching scores of ground truth captions are set as 1.0, logit is the matching score calculated by CoCa.

3.3. Image and sentence encoding

To generate contextualized image representations and sentence representations, we adopt and train a Transformer to encode image inputs and text inputs.

We first get image (**p**icture) embeddings in the **d**ocument denoted as $\boldsymbol{E}^{\mathrm{pd}} = [\boldsymbol{e}_1^{\mathrm{pd}}, \boldsymbol{e}_2^{\mathrm{pd}}, \ldots, \boldsymbol{e}_M^{\mathrm{pd}}]$ and image (**p**icture) embeddings in the **s**ummary denoted as $\boldsymbol{E}^{\mathrm{ps}} = [\boldsymbol{e}_1^{\mathrm{ps}}, \boldsymbol{e}_2^{\mathrm{ps}}, \ldots, \boldsymbol{e}_m^{\mathrm{ps}}]$, using the image encoder of the tuned CoCa model. We get **s**entence embeddings in the **d**ocument denoted as $\boldsymbol{E}^{\mathrm{sd}} = [\boldsymbol{e}_1^{\mathrm{sd}}, \boldsymbol{e}_2^{\mathrm{sd}}, \ldots, \boldsymbol{e}_N^{\mathrm{sd}}]$ and **s**entence embeddings in the **s**ummary denoted as $\boldsymbol{E}^{\mathrm{ss}} = [\boldsymbol{e}_1^{\mathrm{ss}}, \boldsymbol{e}_2^{\mathrm{ss}}, \ldots, \boldsymbol{e}_n^{\mathrm{ss}}]$, using the text encoder of the tuned CoCa model. The tuned CoCa encodes images and sentences into a uniform vector space, and is used for initial image and sentence embedding. Parameters of the tuned CoCa are frozen during diffusion training.

The document sentence embeddings, document image embeddings, summary sentence embeddings, and summary image embeddings are then concatenated and fed into a stacked Transformer followed by a

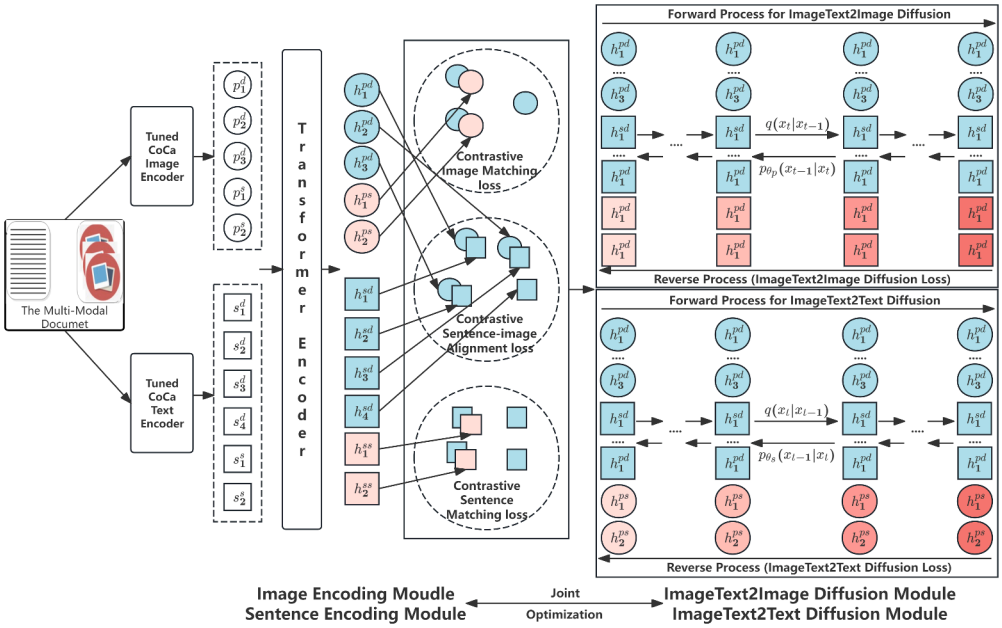


Fig. 1. The overall architecture of DiffuMSMO. The input multi-modal document is first passed to the tuned CoCa model to get image representations and sentence representations. Then the representations are passed to the Transformer encoder, the image diffusion module, and the sentence diffusion module. DiffuMSMO will generate summary sentence representations and summary image representations based on which sentences and images are extracted as the multi-modal summaries.

projection layer to obtain contextualized **d**ocument **s**entence representations $\boldsymbol{H}^{\text{sd}} = [\boldsymbol{h}_1^{\text{sd}}, \boldsymbol{h}_2^{\text{sd}}, \ldots, \boldsymbol{h}_N^{\text{sd}}] \in \mathbb{R}^{N\times h}$, **d**ocument image (**p**icture) representations $\boldsymbol{H}^{\text{pd}} = [\boldsymbol{h}_1^{\text{pd}}, \boldsymbol{h}_2^{\text{pd}}, \ldots, \boldsymbol{h}_M^{\text{pd}}] \in \mathbb{R}^{M\times h}$, **s**ummary **s**entence representations $\boldsymbol{H}^{\text{ss}} = [\boldsymbol{h}_1^{\text{ss}}, \boldsymbol{h}_2^{\text{ss}}, \ldots, \boldsymbol{h}_n^{\text{ss}}] \in \mathbb{R}^{n\times h}$ and **s**ummary image (**p**icture) representations $\boldsymbol{H}^{ps} = [\boldsymbol{h}_1^{\text{ps}}, \boldsymbol{h}_2^{\text{ps}}, \ldots, \boldsymbol{h}_m^{\text{ps}}] \in \mathbb{R}^{m\times h}$ as in Eqs. (3) and (4), where h is the dimension of the representations and [ ] denotes the concatenation operation.

$$\boldsymbol{H}^{\text{sd}}, \boldsymbol{H}^{\text{pd}} = \text{MLP}(\text{Transformer}([\boldsymbol{E}^{\text{sd}}, \boldsymbol{E}^{\text{pd}}])) \tag{3}$$

$$\boldsymbol{H}^{\text{ss}}, \boldsymbol{H}^{\text{ps}} = \text{MLP}(\text{Transformer}([\boldsymbol{E}^{\text{ss}}, \boldsymbol{E}^{\text{ps}}])) \tag{4}$$

The document image representations $\boldsymbol{H}^{\text{pd}}$, the document sentence representations $\boldsymbol{H}^{\text{sd}}$ and the summary sentence representations $\boldsymbol{H}^{\text{ss}}$ are then concatenated as the input $\boldsymbol{H}^{\text{in}}_{\text{it2t}} = [\boldsymbol{H}^{\text{pd}}, \boldsymbol{H}^{\text{sd}}, \boldsymbol{H}^{\text{ss}}] \in \mathbb{R}^{(M+N+n)\times h}$ and will be passed to the ImageText2Text diffusion module. And $\boldsymbol{H}^{\text{pd}}$, $\boldsymbol{H}^{\text{sd}}$ and the summary image representations H^{ps} are then concatenated as the input $\boldsymbol{H}^{\text{in}}_{\text{it2i}} = [\boldsymbol{H}^{\text{pd}}, \boldsymbol{H}^{\text{sd}}, \boldsymbol{H}^{\text{ps}}] \in \mathbb{R}^{(M+N+m)\times h}$ and will be passed to the ImageText2Image diffusion module.

3.3.1. Sentence-image alignment

The parameters of the Transformer for image and sentence encoding are trained to produce alignable image and sentence representations. The loss $\mathcal{L}^{align}$ for training alignable sentence-image (sentence-picture) representations is defined as follows.

Image-to-Sentence Alignment Given an image, we adopt the multi-class supervised contrastive loss (Khosla et al., 2020) to pull the image close to its matching sentence in the source document, and push the image far from its non-matching sentences. We obtain the most matching sentence in the document for the image j in the document by computing the dot product of CoCa embeddings followed by a softmax function as in Eq. (5).

$$\text{align}^{\text{p2s}}(p_j^{\text{d}}) = \arg\max(\text{softmax}(\boldsymbol{e}_j^{\text{pd}} \cdot (\boldsymbol{E}^{\text{sd}})^T)) \tag{5}$$

The image-to-sentence alignment sequence is $\boldsymbol{H}^{\text{p2s}} = [\boldsymbol{h}_1^{\text{sd}}, \boldsymbol{h}_2^{\text{sd}}, \ldots, \boldsymbol{h}_N^{\text{sd}}, \boldsymbol{h}_1^{\text{pd}}, \boldsymbol{h}_2^{\text{pd}}, \ldots, \boldsymbol{h}_M^{\text{pd}}] \in \mathbb{R}^{(N+M)\times h}$, and we suppose $\boldsymbol{h}_i^{\text{p2s}} \in \mathbb{R}^h$ is the ith vector of $\boldsymbol{H}^{\text{p2s}}$. The contrastive label for image-to-sentence $\boldsymbol{y}^{\text{p2s}}$ is defined as in Eq. (6).

$$y_k^{\text{p2s}} = \begin{cases} \min(\{j \mid \text{align}^{\text{p2s}}(p_j^{\text{d}}) = k\}), \text{if } k \le N \text{ and } \text{align}^{\text{p2s}}(p_j^{\text{d}}) = k \\ \min(\{j \mid \text{align}^{\text{p2s}}(p_j^{\text{d}}) = \text{align}^{\text{p2s}}(p_{k-N}^{\text{d}})\}), \text{if } k > N \\ 0, \text{otherwise} \end{cases} \tag{6}$$

where $1 \le j \le M$ and y_k^{p2s} is the kth element of $\boldsymbol{y}^{\text{p2s}}$. Multiple images can be aligned to one sentence, so y_k^{p2s} is set as the minimum index of these images. The contrastive loss for image-to-sentence alignment $\mathcal{L}^{p2s}$ is defined as in Eq. (7)–(9).

$$\mathcal{L}^{\text{p2s}}_{k>N} = -\sum_{\substack{u=1 \\ y_k^{\text{p2s}}=y_u^{\text{p2s}}}}^{N} \log\left(\frac{\exp(\boldsymbol{h}_k^{\text{p2s}} \cdot (\boldsymbol{h}_u^{\text{p2s}})^T/\tau)}{\sum_{v=1}^{N} \exp(\boldsymbol{h}_k^{\text{p2s}} \cdot (\boldsymbol{h}_v^{\text{p2s}})^T/\tau)}\right) \tag{7}$$

$$\mathcal{L}^{\text{p2s}}_{k\le N} = -\sum_{\substack{u=N+1 \\ y_k^{\text{p2s}}=y_u^{\text{p2s}}}}^{N+M} \log\left(\frac{\exp(\boldsymbol{h}_k^{\text{p2s}} \cdot (\boldsymbol{h}_u^{\text{p2s}})^T/\tau)}{\sum_{v=N+1}^{N+M} \exp(\boldsymbol{h}_k^{\text{p2s}} \cdot (\boldsymbol{h}_v^{\text{p2s}})^T/\tau)}\right) \tag{8}$$

$$\mathcal{L}^{\text{p2s}} = \sum_{k\in[1,N+M]} \frac{1}{N_k^{\text{p2s}}} \mathcal{L}_k^{\text{p2s}} \tag{9}$$

where N_k^{p2s} is the total number of sentences (when $k > N$) or images (when $k \le N$) in the document that have the same label y_k^{p2s}, and τ is a temperature hyperparameter. We only count sentence-image alignment parts in Eq. (7) and (8), but do not count sentence-sentence or image-image pairs, because we focus on pulling closer positive sentence-image pairs and pushing farther negative sentence-image pairs. For the image k, the sentences with the same label y_k^{p2s} serve as positive samples, and other sentences serve as negative samples. For the sentence k, the images with the same label y_k^{p2s} serve as positive samples, and other images serve as negative samples. There are no self-comparisons in Eq. (7) and (8).

Sentence-to-Image Alignment Similarly, given a sentence i in the source document, we obtain the most matching image in the source document using the tuned CoCa model as in Eq. (10).

$$\text{align}^{\text{s2p}}(s_i^{\text{d}}) = \arg\max(\text{softmax}(\boldsymbol{e}_i^{\text{sd}} \cdot (\boldsymbol{E}^{\text{pd}})^T)) \tag{10}$$

The sentence-to-image alignment sequence is $\boldsymbol{H}^{\mathrm{s2p}} = [\boldsymbol{h}_1^{\mathrm{pd}}, \boldsymbol{h}_2^{\mathrm{pd}}, \ldots, \boldsymbol{h}_M^{\mathrm{pd}}, \boldsymbol{h}_1^{\mathrm{sd}}, \boldsymbol{h}_2^{\mathrm{sd}}, \ldots, \boldsymbol{h}_N^{\mathrm{sd}}] \in \mathbb{R}^{(M+N)\times h}$. The contrastive label for sentence-to-image $\boldsymbol{y}^{\mathrm{s2p}}$ is defined as in Eq. (11).

$$y_k^{\mathrm{s2p}} = \begin{cases} \min(\{i \mid \mathrm{align}^{\mathrm{s2p}}(s_i^{\mathrm{d}}) = k\}), \text{if } k \le M \text{ and } \mathrm{align}^{\mathrm{s2p}}(s_i^{\mathrm{d}}) = k \\ \min(\{i \mid \mathrm{align}^{\mathrm{s2p}}(s_i^{\mathrm{d}}) = \mathrm{align}^{\mathrm{s2p}}(s_{k-M}^{\mathrm{d}})\}), \text{if } k > M \\ 0, \text{otherwise} \end{cases} \tag{11}$$

where $1 \le i \le N$ and y_k^{s2p} is the kth element of $\boldsymbol{y}^{\mathrm{s2p}}$. Multiple sentences can be aligned to one image, so y_k^{s2p} is set as the minimum index of these sentences. The contrastive loss for sentence-to-image alignment L^{s2p} is computed as in Eqs. (12)–(14).

$$\mathcal{L}_{k>M}^{\mathrm{s2p}} = -\sum_{\substack{u=1 \\ y_k^{\mathrm{s2p}}=y_u^{\mathrm{s2p}}}}^{M} \log\left(\frac{\exp(\boldsymbol{h}_k^{\mathrm{s2p}} \cdot (\boldsymbol{h}_u^{\mathrm{s2p}})^T/\tau)}{\sum_{v=1}^{M} \exp(\boldsymbol{h}_k^{\mathrm{s2p}} \cdot (\boldsymbol{h}_v^{\mathrm{s2p}})^T/\tau)}\right) \tag{12}$$

$$\mathcal{L}_{k\le M}^{\mathrm{s2p}} = -\sum_{\substack{u=M+1 \\ y_k^{\mathrm{s2p}}=y_u^{\mathrm{s2p}}}}^{N+M} \log\left(\frac{\exp(\boldsymbol{h}_k^{\mathrm{s2p}} \cdot (\boldsymbol{h}_u^{\mathrm{s2p}})^T/\tau)}{\sum_{v=M+1}^{N+M} \exp(\boldsymbol{h}_k^{\mathrm{s2p}} \cdot (\boldsymbol{h}_v^{\mathrm{s2p}})^T/\tau)}\right) \tag{13}$$

$$\mathcal{L}^{\mathrm{s2p}} = \sum_{k\in[1,N+M]} \frac{1}{N_k^{\mathrm{s2p}}} \mathcal{L}_k^{\mathrm{s2p}} \tag{14}$$

where N_k^{s2p} is the total number of images (when $k > N$) or sentences (when $k \le N$) in the document that have the same label y_k^{s2p}, and τ is a temperature hyperparameter. As with Image-to-Sentence Alignment, we only count image-sentence and sentence-image pairs, but do not count sentence-sentence or image-image pairs in Eqs. (12) and (13).

The final sentence-image alignment loss is the summation of $\mathcal{L}^{\mathrm{p2s}}$ and $\mathcal{L}^{\mathrm{s2p}}$ as in Eq. (15).

$$\mathcal{L}^{\mathrm{align}} = \mathcal{L}^{\mathrm{p2s}} + \mathcal{L}^{\mathrm{s2p}} \tag{15}$$

3.3.2. Image matching and sentence matching

The parameters of the Transformer are also trained for accurate summary image extraction and summary sentence extraction. As with (Zhang et al., 2023a), we introduce a matching loss and a multiclass supervised contrastive loss by aligning source sentences with summary sentences and aligning source images with summary images.

Image Matching The matching loss for summary image extraction is formally defined as follows. For the summary image representation $\boldsymbol{h}_j^{\mathrm{ps}}$, its matching scores with the source images are computed as the dot product with document representations followed by a softmax function as in Eq. (16). Then the matching loss for summary image extraction $\mathcal{L}_{\mathrm{match}}^{\mathrm{p}}$ is computed as the cross-entropy between the matching score $\hat{\boldsymbol{y}}_{\mathrm{match}}^{\mathrm{p}}$ and the ground truth image summary label $\boldsymbol{y}_{\mathrm{match}}^{\mathrm{p}}$ as in Eq. (17):

$$\hat{\boldsymbol{y}}_{\mathrm{match},j}^{\mathrm{p}} = \mathrm{softmax}(\boldsymbol{h}_j^{\mathrm{ps}} \cdot (\boldsymbol{H}^{\mathrm{pd}})^T) \tag{16}$$

$$\mathcal{L}_{\mathrm{match}}^{\mathrm{p}} = \sum_{j=1}^{m} \mathrm{CrossEntropy}(\boldsymbol{y}_{\mathrm{match},j}^{\mathrm{p}}, \hat{\boldsymbol{y}}_{\mathrm{match},j}^{\mathrm{p}}) \tag{17}$$

where $\hat{\boldsymbol{y}}_{\mathrm{match},j}^{\mathrm{p}}$ is the matching score vector of the jth summary image against the images in the source document, and $\boldsymbol{y}_{\mathrm{match},j}^{\mathrm{p}}$ is the ground truth (ORACLE) label for the jth summary image.

We also add a contrastive loss to make the representations more distinguishable by pulling the summary representation closer to its corresponding source representation while pushing it away from other embeddings.

The image representations are $\boldsymbol{H}^{\mathrm{p}} = [\boldsymbol{h}_1^{\mathrm{pd}}, \boldsymbol{h}_2^{\mathrm{pd}}, \ldots, \boldsymbol{h}_M^{\mathrm{pd}}, \boldsymbol{h}_1^{\mathrm{ps}}, \boldsymbol{h}_2^{\mathrm{ps}}, \ldots, \boldsymbol{h}_m^{\mathrm{ps}}] \in \mathbb{R}^{(M+m)\times h}$, and we suppose $\boldsymbol{h}_j^{\mathrm{p}} \in \mathbb{R}^h$ is the jth vector of $\boldsymbol{H}^{\mathrm{p}}$. The contrastive labels for image matching $\boldsymbol{y}_{\mathrm{contra}}^{\mathrm{p}}$ are defined as in Eq. (18).

$$y_{\mathrm{contra},k}^{\mathrm{p}} = \begin{cases} j, \text{if } k \le M \text{ and } p_j^{\mathrm{s}} = p_k^{\mathrm{d}} \\ j, \text{if } k = M + j \\ 0, \text{otherwise} \end{cases} \tag{18}$$

where $1 \le j \le m$ and $y_{\mathrm{contra},k}^{\mathrm{p}}$ is the kth element of $\boldsymbol{y}_{\mathrm{contra}}^{\mathrm{p}}$. The contrastive loss for image matching $\mathcal{L}_{\mathrm{contra}}^{\mathrm{p}}$ is defined as in Eq. (20). $N_k^{\mathrm{p}} - 1$ in the equation is the total number of image pairs that have the same label $y_{\mathrm{contra},k}^{\mathrm{p}}$ in the document minus the number of self comparisons.

$$\mathcal{L}_{\mathrm{contra},k}^{\mathrm{p}} = -\sum_{\substack{u=1 \\ u\ne k \\ y_k^{\mathrm{p}}=y_u^{\mathrm{p}}}}^{M+m} \log\left(\frac{\exp(\boldsymbol{h}_k^{\mathrm{p}} \cdot (\boldsymbol{h}_u^{\mathrm{p}})^T/\tau)}{\sum_{v=1;v\ne k}^{M+m} \exp(\boldsymbol{h}_k^{\mathrm{p}} \cdot (\boldsymbol{h}_v^{\mathrm{p}})^T/\tau)}\right) \tag{19}$$

$$\mathcal{L}_{\mathrm{contra}}^{\mathrm{p}} = \sum_{k\in[1,M+m]} \frac{1}{N_k^{\mathrm{p}} - 1} \mathcal{L}_{\mathrm{contra},k}^{\mathrm{p}} \tag{20}$$

Eq. (21) is the final loss for image matching.

$$\mathcal{L}^{\mathrm{p}} = \mathcal{L}_{\mathrm{match}}^{\mathrm{p}} + \gamma^{\mathrm{p}} \mathcal{L}_{\mathrm{contra}}^{\mathrm{p}} \tag{21}$$

where γ^{p} is the rescale factor of contrastive image matching loss.

Sentence Matching The matching loss for summary sentence extraction $\mathcal{L}_{\mathrm{match}}^{\mathrm{s}}$ is similar to that of summary image extraction by computing cross-entropy between sentence matching scores and ground truth summary sentence labels. Similarly, the sentence sequence for contrastive sentence matching to enhance sentence diversity is $\boldsymbol{H}^{\mathrm{s}} = [\boldsymbol{h}_1^{\mathrm{sd}}, \boldsymbol{h}_2^{\mathrm{sd}}, \ldots, \boldsymbol{h}_N^{\mathrm{sd}}, \boldsymbol{h}_1^{\mathrm{ss}}, \boldsymbol{h}_2^{\mathrm{ss}}, \ldots, \boldsymbol{h}_n^{\mathrm{ss}}] \in \mathbb{R}^{(N+n)\times h}$, based on which the contrastive loss $\mathcal{L}_{\mathrm{contra}}^{\mathrm{s}}$ is defined to pull the summary sentence representation closer to its corresponding source sentence representation while pushing it away from other sentence embeddings. Eq. (22) is the final loss for sentence matching.

$$\mathcal{L}^{\mathrm{s}} = \mathcal{L}_{\mathrm{match}}^{\mathrm{s}} + \gamma^{\mathrm{s}} \mathcal{L}_{\mathrm{contra}}^{\mathrm{s}} \tag{22}$$

where γ^{s} is the rescale factor of contrastive sentence matching loss.

3.3.3. Training loss for encoding

The final loss to train image and sentence encoding is in Eq. (23).

$$\mathcal{L}^{\mathrm{encoding}} = \mathcal{L}^{\mathrm{p}} + \mathcal{L}^{\mathrm{s}} + \gamma^{\mathrm{align}} \mathcal{L}^{\mathrm{align}} \tag{23}$$

where γ^{align} is the rescale factor of the sentence-image alignment loss.

3.4. Two-line diffusion

As mentioned above, our diffusion model is a two-line diffusion model. The first line is the ImageText2Text (it2t for short) line which aims to generate summary sentence representations based on which summary sentences are extracted, and the second line is the ImageText2Image (it2i for short) line which aims to generate summary image representations based on which summary images are extracted. The input of ImageText2Text is $\boldsymbol{H}_{\mathrm{it2t}}^{\mathrm{in}} = [\boldsymbol{H}^{\mathrm{pd}}, \boldsymbol{H}^{\mathrm{sd}}, \boldsymbol{H}^{\mathrm{ss}}] \in \mathbb{R}^{(M+N+n)\times h}$, and the input of ImageText2Image is $\boldsymbol{H}_{\mathrm{it2i}}^{\mathrm{in}} = [\boldsymbol{H}^{\mathrm{pd}}, \boldsymbol{H}^{\mathrm{sd}}, \boldsymbol{H}^{\mathrm{ps}}] \in \mathbb{R}^{(M+N+m)\times h}$. For either line, we add Gaussian noise gradually through the forward process and fit a Transformer to invert the diffusion in the reverse process.

We first perform one-step Markov transition $q(\boldsymbol{x}_0^{\mathrm{it2t}} | \boldsymbol{H}_{\mathrm{it2t}}^{\mathrm{in}}) = \mathcal{N}(\boldsymbol{H}_{\mathrm{it2t}}^{\mathrm{in}}, \beta_0^{\mathrm{it2t}} \boldsymbol{I})$ and $q(\boldsymbol{x}_0^{\mathrm{it2i}} | \boldsymbol{H}_{\mathrm{it2i}}^{\mathrm{in}}) = \mathcal{N}(\boldsymbol{H}_{\mathrm{it2i}}^{\mathrm{in}}, \beta_0^{\mathrm{it2i}} \boldsymbol{I})$ to get the starting states where $\boldsymbol{x}_0^{\mathrm{it2t}} = [\boldsymbol{x}_0^{\mathrm{pd}}, \boldsymbol{x}_0^{\mathrm{sd}}, \boldsymbol{x}_0^{\mathrm{ss}}] \in \mathbb{R}^{(M+N+n)\times h}$ and $\boldsymbol{x}_0^{\mathrm{it2i}} = [\boldsymbol{x}_0^{\mathrm{pd}}, \boldsymbol{x}_0^{\mathrm{sd}}, \boldsymbol{x}_0^{\mathrm{ps}}] \in \mathbb{R}^{(M+N+m)\times h}$. The initial Markov transition is applied to both document and summary embeddings. We then start the forward process by gradually injecting partial noise to summary representations $\boldsymbol{x}^{\mathrm{ss}} \in \mathbb{R}^{n\times h}$ and $\boldsymbol{x}^{\mathrm{ps}} \in \mathbb{R}^{m\times h}$, and leaving source representations unchanged similar to Gong et al. (2022), enabling the diffusion model to generate conditionally on the source sentences and source images. At step t of the forward process $q(\boldsymbol{x}_t^{\mathrm{ps}} | \boldsymbol{x}_{t-1}^{\mathrm{ps}})$ and

$q(\boldsymbol{x}_t^{\text{ss}}|\boldsymbol{x}_{t-1}^{\text{ss}})$, the noised representations are $\boldsymbol{x}_t^{\text{it2t}}$ and $\boldsymbol{x}_t^{\text{it2i}}$ for the two lines of diffusion models respectively:

$$\boldsymbol{x}_t^{\text{it2t}} = [\boldsymbol{x}_0^{\text{pd}}, \boldsymbol{x}_0^{\text{sd}}, \mathcal{N}(\boldsymbol{x}_t^{\text{ss}}; \sqrt{1-\beta_t^{\text{it2t}}}\boldsymbol{x}_{t-1}^{\text{ss}}, \beta_t^{\text{it2t}}\boldsymbol{I})] \tag{24}$$

$$\boldsymbol{x}_t^{\text{it2i}} = [\boldsymbol{x}_0^{\text{pd}}, \boldsymbol{x}_0^{\text{sd}}, \mathcal{N}(\boldsymbol{x}_t^{\text{ps}}; \sqrt{1-\beta_t^{\text{it2i}}}\boldsymbol{x}_{t-1}^{\text{ps}}, \beta_t^{\text{it2i}}\boldsymbol{I})] \tag{25}$$

Once the partially noised representations are acquired, the backward process is conducted to remove the noise of summary representations given the condition of the summary sentence representations or image representations of the previous step:

$$P_{\theta_{\text{it2t}}}(\boldsymbol{x}_{t-1}^{\text{ss}}|\boldsymbol{x}_t^{\text{it2t}}) = \mathcal{N}(\boldsymbol{x}_{t-1}^{\text{ss}}; \mu_{\theta_{\text{it2t}}}(\boldsymbol{x}_t^{\text{it2t}}, t), \sigma^2_{\theta_{\text{it2t}}}(t)\boldsymbol{I}) \tag{26}$$

$$P_{\theta_{\text{it2i}}}(\boldsymbol{x}_{t-1}^{\text{ps}}|\boldsymbol{x}_t^{\text{it2i}}) = \mathcal{N}(\boldsymbol{x}_{t-1}^{\text{ps}}; \mu_{\theta_{\text{it2i}}}(\boldsymbol{x}_t^{\text{it2i}}, t), \sigma^2_{\theta_{\text{it2i}}}(t)\boldsymbol{I}) \tag{27}$$

where $\mu_{\theta_{\text{it2t}}}(\cdot)$, $\sigma^2_{\theta_{\text{it2t}}}(\cdot)$, $\mu_{\theta_{\text{it2i}}}(\cdot)$, $\sigma^2_{\theta_{\text{it2i}}}(\cdot)$ are parameterized models to predict the mean and standard variation at diffusion step $t-1$, and θ_{it2t} and θ_{it2i} are the parameters to be optimized. The final outputs of the two-line diffusion model after T reverse steps are the generated summary image representations $\widetilde{\boldsymbol{H}}_0^{\text{ps}} = [\widetilde{\boldsymbol{h}}_1^{\text{ps}}, \widetilde{\boldsymbol{h}}_2^{\text{ps}}, \ldots, \widetilde{\boldsymbol{h}}_m^{\text{ps}}] \in \mathbb{R}^{m\times h}$ and summary sentence representations $\widetilde{\boldsymbol{H}}_0^{\text{ss}} = [\widetilde{\boldsymbol{h}}_1^{\text{ss}}, \widetilde{\boldsymbol{h}}_2^{\text{ss}}, \ldots, \widetilde{\boldsymbol{h}}_n^{\text{ss}}] \in \mathbb{R}^{n\times h}$. We optimize the diffusion generation module with diffusion loss $\mathcal{L}_{\text{diffusion}}$ defined as:

$$\begin{aligned}\mathcal{L}^{\text{diffusion}} = &\sum_{t=2}^{T} \|\boldsymbol{x}_0^{\text{it2t}} - \widetilde{f}_{\theta_{\text{it2t}}}(\boldsymbol{x}_t^{\text{it2t}}, t)\|^2 + \\ &\|\boldsymbol{H}_{\text{it2t}}^{\text{in}} - \widetilde{f}_{\theta_{\text{it2t}}}(\boldsymbol{x}_1^{\text{it2t}}, 1)\|^2 + \mathcal{R}(\boldsymbol{x}_0^{\text{it2t}}) + \\ &\sum_{t=2}^{T} \|\boldsymbol{x}_0^{\text{it2i}} - \widetilde{f}_{\theta_{\text{it2i}}}(\boldsymbol{x}_t^{\text{it2i}}, t)\|^2 + \\ &\|\boldsymbol{H}_{\text{it2i}}^{\text{in}} - \widetilde{f}_{\theta_{\text{it2i}}}(\boldsymbol{x}_1^{\text{it2i}}, 1)\|^2 + \mathcal{R}(\boldsymbol{x}_0^{\text{it2i}})\end{aligned} \tag{28}$$

where $\widetilde{f}_{\theta_{\text{it2t}}}(\boldsymbol{x}_t^{\text{it2t}}, t)$ and $\widetilde{f}_{\theta_{\text{it2i}}}(\boldsymbol{x}_t^{it2i}, t)$ are the reconstructed $\boldsymbol{x}_0^{\text{it2t}}$ and $\boldsymbol{x}_0^{\text{it2i}}$ at step t, and $\mathcal{R}(\boldsymbol{x}_0^{\text{it2t}})$ and $\mathcal{R}(\boldsymbol{x}_0^{\text{it2i}})$ are L2 regularization terms. $\mathcal{L}^{\text{diffusion}}$ is the summation of diffusion loss of two lines of diffusion models.

3.5. Optimization and inference

The overall objective function is:

$$\mathcal{L} = \mathcal{L}^{\text{encoding}} + \mathcal{L}^{\text{diffusion}} \tag{29}$$

In the inference stage, which involves only the reverse process, DiffuMSMO first encodes source images and source sentences of the multi-modal document into $\boldsymbol{H}^{\text{pd}}$ and $\boldsymbol{H}^{\text{sd}}$, and performs a one-step Markov transition. The model then randomly samples m noise vectors from Gaussian noise to construct noised summary image representations $\boldsymbol{x}_T^{\text{ps}} \sim \mathcal{N}(0, \boldsymbol{I})$ and samples n noise vectors to construct noised summary sentence representations $\boldsymbol{x}_T^{\text{ss}} \sim \mathcal{N}(0, \boldsymbol{I})$ at step T. During the reverse process, DiffuMSMO gradually denoises $\boldsymbol{x}_T^{\text{it2i}} = [\boldsymbol{x}_0^{\text{pd}}, \boldsymbol{x}_0^{\text{sd}}, \boldsymbol{x}_T^{\text{ps}}]$ and $\boldsymbol{x}_T^{\text{it2t}} = [\boldsymbol{x}_0^{\text{pd}}, \boldsymbol{x}_0^{\text{sd}}, \boldsymbol{x}_T^{\text{ss}}]$ to obtain the predicted embeddings of the summary images $\widetilde{\boldsymbol{H}}_0^{\text{ps}} = [\widetilde{\boldsymbol{h}}_1^{\text{ps}}, \widetilde{\boldsymbol{h}}_2^{\text{ps}}, \ldots, \widetilde{\boldsymbol{h}}_m^{\text{ps}}]$ and summary sentences $\widetilde{\boldsymbol{H}}_0^{\text{ss}} = [\widetilde{\boldsymbol{h}}_1^{\text{ss}}, \widetilde{\boldsymbol{h}}_2^{\text{ss}}, \ldots, \widetilde{\boldsymbol{h}}_n^{\text{ss}}]$. Finally, based on the similarity between $\widetilde{\boldsymbol{h}}_j^{\text{ps}}$ and source image representations $\boldsymbol{H}^{\text{pd}}$, DiffuMSMO selects the most closely matching image as its corresponding summary image, and based on the similarity between $\widetilde{\boldsymbol{h}}_i^{\text{ss}}$ and source sentence representations $\boldsymbol{H}^{\text{sd}}$, DiffuMSMO selects the most closely matching sentences as its corresponding summary sentence. When an image or a sentence is selected, it is marked as selected and will not be considered for next selection round. The selection procedure will stop if all the sentences or images are marked selected or the selected number reaches a threshold. The sentence-image alignment scores are computed to align summary sentences and summary images.

Table 1
Statistics of the MSMO dataset. The values with * are for the greedily created extractive summaries as described in Section 3.1.

Statistics	Train	Valid	Test
Sample Number	293965	10355	10261
Avg.Tokens.Source	720.87	766.08	730.80
Avg.Tokens.Target	70.12	70.02	72.16
Avg.Images.Source	6.56	6.91	6.67
Avg.Images.Target	2.54*	2.60*	3.68
Avg.Sents.Source	23.78	25.35	24.02
Avg.Sents.Target	3.03*	3.02*	4.02

4. Experiments

4.1. Dataset

For multi-modal summarization with multi-modal output, the MSMO dataset is used which is introduced in Zhu et al. (2018). This is the first and only large-scale English corpus specifically curated for this task. The MSMO dataset comprises a collection of online news articles collected from the DailyMail website, each accompanied by several images and manually-written highlights that serve as the reference summary. The statistics of the dataset are illustrated in Table 1. The average target image number and sentence number for the greedily created extractive summaries are also reported in the table. Within the test set, summary images are manually annotated to provide a pictorial reference. The ORACLE image labels of the train and valid set are obtained as introduced in Section 3.1

The ROUGE metric (Lin, 2004) which quantifies the degree of lexical correspondence between the produced sentences and the reference summaries is employed to evaluate the generated textual summaries. All the ROUGE scores in this paper refer to the F-1 scores calculated by the rouge155 script. In addition, the metric of image precision (IP) is introduced to evaluate the quality of the selected image as calculated in Eq. (30) (Zhu et al., 2018).

$$\text{IP} = \frac{\left|\text{ref}_{\text{img}} \cap \text{sel}_{\text{img}}\right|}{\left|\text{sel}_{\text{img}}\right|} \tag{30}$$

where ref_{img} and sel_{img} are the reference images and the selected images respectively.

4.2. Implementations

We fine-tune the pre-trained CoCa in the news image captioning dataset NYTimes800K for the news domain. We distill knowledge in Sentence-Bert (Reimers & Gurevych, 2019) by collecting sentences similar to ground-truth captions from source documents as the complement captions. The fine-tuned CoCa is used to get initial sentence embeddings and image embeddings. The weight η for CoCa fine-tuning is set as 0.75. The sentence and image encoder module is constructed using 8 stacked transformer encoders. Another 12 stacked transformer encoders are used for the diffusion module. The temperature hyperparameter τ is set as 0.07. The hidden size of DiffuMSMO is 768. The maximum diffusion step T is set to 500, and weight parameters $\gamma^{\text{p}} = 0.001$, $\gamma^{\text{s}} = 0.001$ and $\gamma^{\text{align}} = 0.001$. For optimization, we utilize the AdamW optimizer (Kingma & Ba, 2014) with a learning rate of $1e-5$ and a dropout rate of 0.1. The batch size is 64, the max number of sentences in document is 45, and the max number of images in document is 10. The max number of sentences in summaries is set as 3 and the max number of images in summaries is set as 3, according to the average target sentence number and average target image number reported in Table 1. We randomly select five seeds for training our model and report the average score. The

model is trained on GeForce RTX 3090 GPUs. Our code will be available on GitHub.[1]

4.3. Baselines

To demonstrate the efficacy of the proposed model, we conduct comparative analyses with extant methodologies in both text-based and multi-modal summarization domains.

- BertSum (Liu & Lapata, 2019) adopts a general framework for both extractive and abstractive text summarization, with the encoder based on BERT (Devlin et al., 2019). Among its several variants, BertAbs and BertExt are for abstractive and extractive summarization respectively.
- BART (Lewis et al., 2020) employs a sequence-to-sequence framework with significant applicability in the domain of text summarization, and is pre-trained on several large summarization datasets.
- ATG/ATL/HAN uses the pointer-generator network (See et al., 2017) and the multi-modal attention mechanism, with variants reflecting different image representation approaches for attention operations.
- MOF^{RR} (Zhu et al., 2020) ranks images via ROUGE score comparison of captions to textual reference, and forms a visual reference. MOF$^{RR}_{enc}$ and MOF$^{RR}_{dec}$ are two variants of incorporating different hidden states into image discriminator.
- UniMS (Zhang et al., 2022b) re-pretrains and fine-tunes the BART model by merging textual and visual data to BART encoder, constructing a multi-modal representation. And it then employs a visually guided decoder to integrate textual and visual modalities to guide abstractive text generation.
- EGMS (Zhang et al., 2024) re-trains and fine-tunes the BART model by incorporating entity-specific information into BART and introducing a pair of multi-modal encoders with shared weights to concurrently process text-image and entity-image information. And a gating mechanism is used to fuse the visual representations.
- DiffuSum (Zhang et al., 2023a) is a diffusion-based extractive text summarization model. It generates the desired summary sentence representations with diffusion models and extracts sentences based on sentence representation matching.
- HGMS(Graphormer)/HGMS(GAT) (Jiang & Chen, 2025) is heterogeneous graph-based model for extractive multimodal summarization (HGMS) designed to efficiently leverage intermodal relationships within multimodal data. A heterogeneous graph is constructed based on the relationships between modalities, containing nodes for words, sentences and images. HGMS(Graphormer) uses an enhanced Graphormer to encode nodes and HGMS(GAT) uses GAT to encode nodes for extractive multi-modal summarization.

4.4. Comparisons

Experimental results on the MSMO dataset are shown in Table 2. The first block in the table shows the scores of the ORACLE extractive text summaries and the ORACLE extractive image summaries greedily created as described in Section 3.1 for the test set. The scores for ORACLE extractive text summaries are computed against the human-written text summaries and can serve as the upper bound for text summary evaluation. The scores for ORACLE extractive image summaries are computed against manually annotated image labels in the test data, and can serve as a reference-informed heuristic benchmark. The second block in the table contains three abstractive text summarization methods BertAbs, BertExtAbs and Bart. The third block includes seven baseline abstractive multi-modal summarization methods. The fourth block includes two baseline extractive text summarization methods. The fifth block includes two baseline extractive multi-modal summarization methods and our proposed method DiffuMSMO.

[1] https://github.com/jingqiangchen/DiffuMSMO

Table 2
Experimental results for multimodal summarization on the MSMO dataset. Results marked by * are taken from respective papers and (Zhang et al., 2024). The bold values of R-1 and R-L indicate statistically significant improvements (p-value $= 1.9 \times 10^{-5}$ for R-1 and 1.0×10^{-8} for R-L) against HGMS(Graphormer) via a one-sided paired t-test (paired per document) on the full test set.

Model	R-1	R-2	R-L	IP
ORACLE Extractive				
ORACLE Text Extractive	57.47	31.85	53.41	–
ORACLE Image Extractive	–	–	–	91.15
Text Abstractive				
BertAbs*	39.02	18.17	33.20	–
BertExtAbs*	39.88	18.77	38.36	–
Bart*	42.93	19.95	39.97	–
Multi-Modal Abstractive				
ATG*	40.63	18.12	37.53	59.28
ATL*	40.86	18.27	37.75	62.44
MOF^{RR*}_{enc}	40.82	18.30	37.70	61.83
MOF^{RR*}_{dec}	41.05	18.29	37.74	62.63
HAN*	41.20	18.33	37.80	65.45
UniMS*	42.94	20.50	40.96	69.38
EGMS*	44.47	21.20	41.43	75.81
Text Extractive				
BertExt	43.46	19.54	39.79	–
DiffuSum	44.58	20.74	40.93	–
Multi-Modal Extractive				
HGMS(GAT)*	44.68	21.34	40.83	78.95
HGMS(Graphormer)*	44.96	21.58	41.08	79.72
DiffuMSMO	**45.37**	21.42	**41.68**	79.05

The results are analyzed as follows. 1) Extractive methods achieve higher scores for both text summarization and multi-modal summarization. Abstractive methods suffer from hallucinations that lower the ROUGE scores. Currently most research on multi-modal summarization focuses on abstractive methods with development of pre-trained generative methods. Nevertheless, extractive methods have advantages in ROUGE scores according to Table 2. 2) DiffuMSMO outperforms DiffuSum in all metrics. The key difference between the two models is that DiffuMSMO considers images in the diffusion-based summarization model through adding images into the input and adding the image diffusion module and image-sentence alignment module into the model. This indicates that considering images in the diffusion-based summarization model can improve multi-modal summarization. 3) DiffuMSMO can achieve competitive performance compared with other extractive multi-modal summarization methods such as HGMS. Especially, DiffuMSMO outperforms the Graphormer-based HGMS model in R-1 and R-L scores with statistically significant improvements at p-value $= 1.9 \times 10^{-5}$ and p-value $= 1.0 \times 10^{-8}$ respectively using a one-sided paired Student's t-test (paired per document) on the full test set. The possible reason is as follows. HGMS utilizes multi-modal relationships in documents by constructing sentence-word-image graphs and the multi-modal Graphormer is proposed to encode nodes for direct sentence and image extraction. As with HGMS, DiffuMSMO also utilizes sentence-image relationships through the sentence-image alignment module in the diffusion model. Multi-modal relationships are important to improve multi-modal summarization. The IP score of DiffuMSMO is lower, possibly because the diffused image encodings are not good enough to extract salient images as HGMS(Graphormer). Nevertheless, the scores are still higher than that of HGMS(GAT).

4.5. Ablation study

To understand the performance of DiffuMSMO, we perform an ablation study by removing or changing model components of the encod-

Table 3
Results of ablation study.

Model	R-1	R-2	R-L	IP	IP.s2i
DiffuMSMO	45.37	21.42	41.68	79.05	75.16
w/o align loss	45.21	21.26	41.52	78.49	75.10
w/o contra loss	45.16	21.26	41.50	78.88	75.03
w/o align&contra loss	44.63	20.74	40.96	78.47	74.78
w/o it2i diffusion line	45.24	21.31	41.54	55.09	75.06
w/o it2t diffusion line	26.26	7.60	23.66	73.33	–
w merged diffusion line	35.59	13.01	32.17	69.30	–
w direct classifier	38.12	15.31	34.63	65.41	–
w transformer projection	38.08	15.52	34.60	64.57	–
w/o CoCa-finetune	45.25	21.38	41.58	79.00	75.02

ing module, the it2i diffusion line, the it2t diffusion line, the summary image selection method, and CoCa finetuning, and show the results in Table 3.

4.5.1. Effect of alignment and contrastive losses

For comparison, we place the performance of DiffuMSMO in the first row. The second row shows that removing the sentence-image alignment loss degrades the performance, indicating that alignment loss is effective for the encoding module. The third row shows that removing the sentence and image contrastive matching loss degrades performance, indicating the contrastive loss is also effective for the encoding module. Both the alignment loss and contrastive loss benefit the overall model performance. The fourth row shows that the performance degrades dramatically when removing both losses. The matching loss is critical to the model. These results prove the effectiveness of jointly training an image and sentence encoder that produces accurate and diverse representations with the generation module.

4.5.2. Effect of summary image selection methods

For summary image selection, we use the extracted summary sentences to select most similar images from the document based on sentence-image similarity computed by CoCa. The results are shown in the sixth column of in Table 3 named as the IP.s2i scores, i.e. IP scores of the summary-sentence-to-image method. In contrast, the image summary for the IP scores in the fifth column are created by matching the images with the image representations generated by the it2i diffusion line. The IP.s2i scores of models with the it2i diffusion line are lower than their corresponding IP scores, and the IP.s2i score of model without the it2i diffusion line is much higher than their corresponding IP score. This indicates that the it2i diffusion is effective for summary image selection.

4.5.3. Effect of diffusion lines

To evaluate the effectiveness of the diffusion lines for text summary generation and summary image selection, we try to 1) remove either line, 2) merge the lines, or 3) replace the diffusion lines with transformer projection or direct classifier. The results are shown in Table 3.

For DiffuMSMO without the it2i diffusion line, we remove the it2i part of diffusion loss in Eq. (28). In this case, it2i diffusion is not trained by the it2i diffusion loss, but is still used for inference. The IP score of this model is much lower, showing that the it2i line is important for image summary representation generation. The ROUGE scores for this model are lower than that of DiffuMSMO, showing that adding the it2i diffusion loss is also effective to text summarization.

For DiffuMSMO without the it2t diffusion line, we remove the it2t part of diffusion loss in Eq. (28). In this case, it2t diffusion is not trained by the it2t diffusion loss, but is still used for inference. The IP score of this model is lower than that of DiffuMSMO, showing that removing the it2t diffusion loss also decreases generation of image summary representations. The ROUGE scores of this model are far lower than that of DiffuMSMO, indicating that the it2t diffusion loss is important for generation of text summary representations.

For DiffuMSMO with merged diffusion line, we merge the two diffusion lines into one line by diffusing sentence representations and image representations indiscriminately using one diffusion loss. The ROUGE scores are lower than that of DiffuMSMO, and the IP score of this model is also lower than that of DiffuMSMO, indicating that merging the two lines into one line decreases both summary sentence representation generation and summary image representation generation.

For DiffuMSMO with direct classifier, we replace the diffusion lines with a sentence classifier and an image classifier. Eq. (31) computes the probability that the ith sentence in the source document is classified as a summary sentence, and Eq. (32) computes the probability that the jth image in the source document is classified as a summary image.

$$P(\hat{y}_i^{\text{sd}} = 1|\boldsymbol{h}_i^{\text{sd}}) = \text{sigmoid}(\text{MLP}(\boldsymbol{h}_i^{\text{sd}})) \tag{31}$$

$$P(\hat{y}_j^{\text{pd}} = 1|\boldsymbol{h}_j^{\text{pd}}) = \text{sigmoid}(\text{MLP}(\boldsymbol{h}_j^{\text{pd}})) \tag{32}$$

where $\boldsymbol{h}_i^{\text{sd}}$ and $\boldsymbol{h}_j^{\text{pd}}$ are the representations computed by Eq. (3). Then the negative log-likelihood loss is computed against the ORACLE labels for training. For inference, summary sentences and summary images are directly extracted using the classifiers. As shown in Table 3, the scores of this model are lower than that of DiffuMSMO. Diffusion models are more suitable than the direct classifier for DiffuMSMO to extract summary sentences and summary images.

For DiffuMSMO with transformer projection, we replace the diffusion lines with a Transformer-based sentence projection and a Transformer-based image projection as in Eqs. (33) and (34).

$$[\overline{\boldsymbol{h}}_1^{\text{sd}}, \ldots, \overline{\boldsymbol{h}}_N^{\text{sd}}, \overline{\boldsymbol{h}}_1^{\text{ss}}, \ldots, \overline{\boldsymbol{h}}_n^{\text{ss}}] = \text{Transformer}_{\text{sentproj}}([\boldsymbol{h}_1^{\text{sd}}, \ldots, \boldsymbol{h}_N^{\text{sd}}, \boldsymbol{r}_1^{\text{ss}}, \ldots, \boldsymbol{r}_n^{\text{ss}}]) \tag{33}$$

$$[\overline{\boldsymbol{h}}_1^{\text{pd}}, \ldots, \overline{\boldsymbol{h}}_M^{\text{pd}}, \overline{\boldsymbol{h}}_1^{\text{ps}}, \ldots, \overline{\boldsymbol{h}}_m^{\text{ps}}] = \text{Transformer}_{\text{imgproj}}([\boldsymbol{h}_1^{\text{pd}}, \ldots, \boldsymbol{h}_M^{\text{pd}}, \boldsymbol{r}_1^{\text{ps}}, \ldots, \boldsymbol{r}_m^{\text{ps}}]) \tag{34}$$

where $\boldsymbol{r}_1^{\text{ss}}, \ldots, \boldsymbol{r}_n^{\text{ss}}$ and $\boldsymbol{r}_1^{\text{ps}}, \ldots, \boldsymbol{r}_m^{\text{ps}}$ are randomly initialized vectors whose sizes are the same as $\boldsymbol{h}_1^{\text{sd}}$ and $\boldsymbol{h}_1^{\text{pd}}$. $\text{Transformer}_{\text{sentproj}}$ and $\text{Transformer}_{\text{imgproj}}$ are the Transformer Encoder for sentence projection and image projection. The resultant $\overline{\boldsymbol{h}}_1^{\text{ss}}, \ldots, \overline{\boldsymbol{h}}_n^{\text{ss}}$ and $\overline{\boldsymbol{h}}_1^{\text{ps}}, \ldots, \overline{\boldsymbol{h}}_m^{\text{ps}}$ accumulate information of the sentence presentations and image representation, and are used as the summary sentence representations and summary image representations instead of the diffusion-based counterparts. As shown in Table 3, the scores of this model are lower than that of DiffuMSMO. Diffusion models are more suitable than the transformer projection for DiffuMSMO to generate summary sentence representations and summary image representations based on which sentences and images are extracted.

4.5.4. Effect of CoCa for the proposed model

To show the effect of CoCa finetuning, w/o CoCa-finetune in Table 3 is the model that replaces the finetuned News-Domain-Specific CoCa with the original CoCa model (Yu et al., 2022). In this case, the untuned CoCa is only used to encode sentences and images to get E^{sd}, E^{ss}, E^{pd}, E^{ps} as in Section 3.3, and the ORACLE image summary labels used in training are constructed using the fine-tuned CoCa model. It shows that performance drops compared with the original DiffuMSMO.

In addition, CoCa is used to create image labels for training by greedily matching images with human-written text summaries as introduced in Section 3.1. And the sentence labels are also created by matching sentences with the text summaries automatically. Both image labels and sentence labels are created based on the text summaries. The it2i diffusion line and the it2t diffusion line can also enhance each other via the correlated labels. The performance of the model decreases when the it2i line or the it2t line is removed as shown in Table 3.

Table 4
Statistics of sub-datasets of MSMO.

Sub-Dataset	Train	Valid	Test
Num.Doc.Img ≤ 8	231353	7871	8014
Num.Doc.Img ≤ 6	195335	6543	6819
Num.Doc.Img ≤ 4	132457	4197	4750

Table 5
Results of DiffuMSMO and DiffuSum in different sub-datasets.

Sub-Dataset	Model	R-1	R-2	R-L	IP
Num.Doc.Img ≤ 8	DiffuMSMO	45.28	21.34	41.60	79.37
	DiffuSum	44.61	20.73	41.01	–
Num.Doc.Img ≤ 6	DiffuMSMO	45.57	21.67	41.69	78.71
	DiffuSum	44.80	21.05	41.11	–
Num.Doc.Img ≤ 4	DiffuMSMO	45.50	21.72	41.73	78.20
	DiffuSum	44.74	21.13	41.18	–

Table 6
Model performance with different γ^{align} values.

γ^{align}	R-1	R-2	R-L	IP
0.01	45.27	21.31	41.46	79.02
0.001	**45.37**	**21.42**	**41.68**	**79.05**
0.0001	45.01	21.14	41.33	78.63

Table 7
Model performance with different diffusion steps.

T	R-1	R-2	R-L	IP
1000	45.29	21.30	41.60	78.94
500	**45.37**	**21.42**	**41.68**	**79.05**
300	45.33	21.38	41.66	79.03
200	45.25	21.38	41.58	79.00
100	45.21	21.33	41.55	79.00

Table 8
Model performance with different settings of numbers of summary sentences and summary images.

Number settings	R-1	R-2	R-L	IP
N.S=2 and N.I=2	42.82	19.60	38.86	81.79
N.S=3 and N.I=3	**45.37**	**21.42**	**41.68**	**79.05**
N.S=4 and N.I=4	45.01	21.05	41.33	76.38

Table 9
Computational efficiency of the models.

Model	Num.Paras	Batch.Size	GPU.Mem
DiffuMSMO in training	219M	64	15G
DiffuMSMO in inference	219M	50	5.86G
DiffuSum in training	133M	64	13G

4.6. Within-dataset sensitivity analyses

Since the MSMO dataset is the only large-scale English corpus specifically curated for this task, we create sub-datasets with different numbers of images from the MSMO dataset to test the proposed method in different situations. We choose to create sub-datasets with different image numbers because we want to show the influence of images for DiffuMSMO. Table 4 shows the sample number of train, valid and test split of sub-datasets under different image number settings of images in source documents (Num.Doc.Img for short), where Num.Doc.Img ≤ k means the number of images in the documents of the sub-dataset is less than or equal to k.

Table 5 shows the results of DiffuMSMO and DiffuSum in the sub-datasets. Firstly, for all the sub-datasets, DiffuMSMO outperforms DiffuSum according to ROUGE scores, indicating that considering images in the summarization model can improve text summarization. Secondly, the IP score of DiffuMSMO is highest when setting Num.Doc.Img ≤ 8, but decreases when setting Num.Doc.Img ≤ 6 or Num.Doc.Img ≤ 4. The number of images in source documents affects summary image selection of the proposed model.

4.7. Impact of hyperparameters

4.7.1. Impact of sentence-image alignment weight

We evaluate DiffuMSMO with the sentence-image alignment weight γ^{align} values of 0.0001, 0.001 and 0.01, as shown in Table 6. Performance drops significantly when γ^{align} deviates from 0.001. For $\gamma^{\text{align}} = 0.0001$ the R-1 score decreases by 0.36 points, indicating that adding sentence-image alignment loss greatly improves performance. And for $\gamma^{\text{align}} = 0.01$ the R-1 score decreases by 0.1 points, indicating that the weight of sentence-image alignment loss is not the bigger the better. Setting $\gamma^{\text{align}} = 0.001$ is appropriate to balance the influence of sentence-image alignment loss.

4.7.2. Impact of diffusion steps

This section compares DiffuMSMO with different diffusion steps, as shown in Table 7. When $T < 500$, the performance of DiffuMSMO improves with increasing T. And when $T > 500$, the performance decreases with increasing T. It is argued that the noise injected into the forward pass cannot be fully removed if the steps are too small, and the model will introduce too much noise to recover if the steps are too big (Zhang et al., 2023a). This also holds for DiffuMSMO.

4.7.3. Impact of numbers of sentences and images in summaries

Since the average numbers of summary sentences and summary images in the train set are 3.03 and 2.54 respectively according to Table 1, we set both the max extracted sentence number (N.S) and the max extracted image number (N.I) as 3 in the experiment. To show effect of the number settings, we also carry out ablations by setting N.S and N.I as 2 or 4, and the results are shown in Table 8. For the numbers setting as 2, the ROUGE scores decrease, because the reported ROUGE scores are the F-measure scores and 2 summary sentences cannot cover information of reference summaries. And the IP score increases, because the IP score is a precision measure and the top-2 selected summary images are more precise than top-3 ones. For the numbers setting as 4, the ROUGE scores decrease, and the IP scores of this setting also decrease.

4.8. Computational efficiency

Table 9 lists the number of trainable parameters and GPU memory consumption of DiffuMSMO and DiffuSum. Memory values are reported under their respective batch sizes.

The number of parameters of DiffuMSMO is approximately 86M larger than that of DiffuSum, since the additional it2i diffusion branch introduces extra parameters. DiffuMSMO consumes 15 GB of GPU memory with a batch size of 64. Training takes roughly 35 h per epoch on a single GeForce RTX 3090 GPU. This runtime includes CoCa embedding computation and can be reduced by precomputing and caching image and sentence embeddings from CoCa; we train the model for 10 epochs in total. Separately, fine-tuning CoCa on the NYT800K news captioning dataset takes around 24 h on one GPU. For inference on the full test set, DiffuMSMO takes approximately four hours on a single GPU with 500 diffusion steps, yielding an average per-document inference latency of 1.40 s. Overall, the model achieves competitive summarization performance at the cost of non-negligible computational overhead, which should be considered for practical deployment.

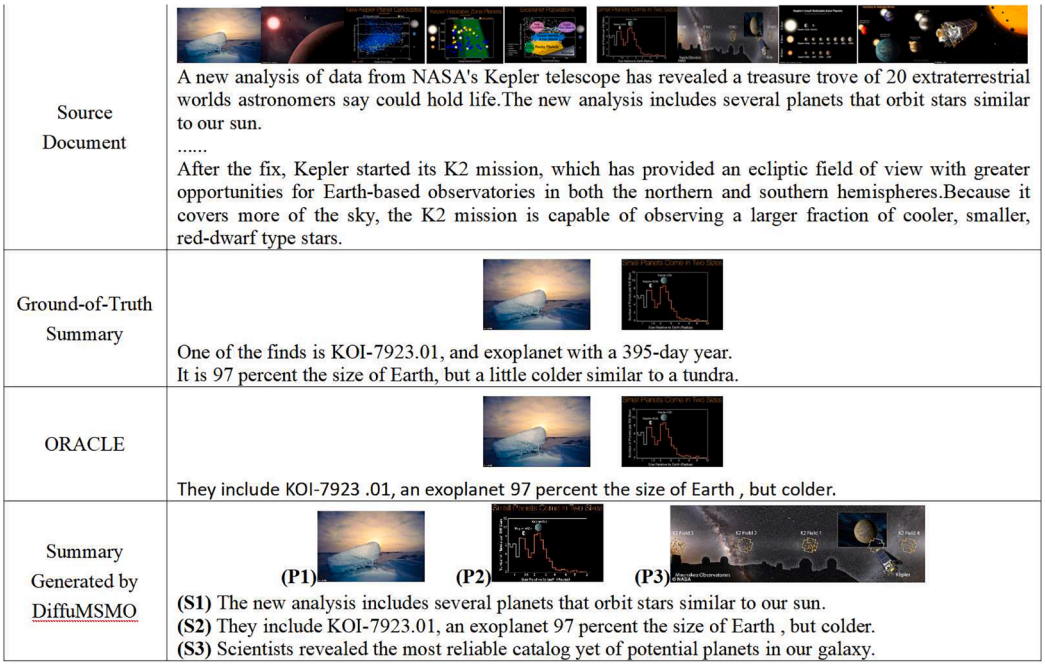


Fig. 2. An example of case study.

Table 10
Salience scores of the extracted sentences and images.

No.	Score	No.	Score
S1	0.1162	**P1**	0.3749
S2	0.1760	**P2**	0.3054
S3	0.0080	**P3**	0.2761

Table 11
Image-to-sentence alignment scores.

	S1	S2	S3
P1	0.0083	**0.7458**	0.2459
P2	0.0600	**0.7307**	0.2093
P3	0.0489	0.2092	**0.7418**

4.9. Case study

We visualize a case of a news article taken from the test data to evaluate the multi-modal summaries generated by DiffuMSMO and the text summary generated by DiffuSum, as shown in Fig. 2. The figure consists of three parts. The first part is the source document, containing 42 sentences and 10 images. The second part is the ground-truth abstractive summary manually created, containing 2 sentences and 2 images. The third part is the ORACLE extractive summary created greedily based on the human-written summary as described in Section 3.1. The ORACLE summary contains 1 sentence and 2 images, and serves as summary sentence labels and summary image labels for DiffuMSMO. The fourth part is the summary generated by DiffuMSMO, containing 3 sentences and 3 images.

The summary generated by DiffuMSMO contains more sentences and images than ORACLE summary, because we force the model to extract at least and at most 3 summary sentences and 3 summary images from the source document. Table 10 shows the salience scores of the sentences and images, computed as the similarity between the sentence/image representations and the diffused sentence/image embeddings. Table 11 shows the image-to-sentence alignment scores computed by the tuned CoCa model. 1) As Table 10 shows, the summary sentence numbered S2 obtains the highest score among all the sentences, which is just the ORACLE summary sentence. Despite the sentences S1 and S3 obtains lower scores, they are also about the planets similar to Earth, not deviating too much from the topic of the sentence S2. 2) The summary images numbered P1 and P2 obtain the highest scores among all the images, which are just the ORACLE summary images. Image P3 relates to the "Kepler mission" and does not deviate from the summary topic. This implies that the diffused sentence and image representations generated by DiffuMSMO are faithful to the ORACLE summary. 3) As Table 11 shows, the images P1 and P2 with highest salience scores are aligned to the sentence S2 with the highest salience score, and the image P3 with lower salience is aligned to the sentence S3 with lower salience score. This implies that sentence diffusion and image diffusion can promote each other. This case study illustrates successful multi-modal selection, while ablations provide the primary evidence for the benefits from joint training.

5. Conclusion

This paper introduces DiffuMSMO, a novel diffusion-based model for the extractive MSMO task. We fine-tune a news-domain-specific CoCa model for initial image and sentence encoding to bridge the gap between textual and visual embeddings. Additionally, we introduce a multi-modal encoder trained with sentence-image alignment loss, sentence matching loss and image matching loss. A two-line multi-modal diffusion model consisting of the ImageText2Image diffusion line and the ImageText2Text diffusion line is proposed to generate summary image representations and summary sentence representations, which are used to extract sentences and images from source documents. Experiments on the MSMO dataset validate the effectiveness of our approach.

Limitations

Despite the promising performance of DiffuMSMO, it still has the following limitations. First, DiffuMSMO is evaluated only on a single multi-

modal summarization dataset. Further investigation is required to assess its generalization by applying the model to additional datasets. Second, the fine-tuned CoCa is used for both image pseudo-label construction and representation initialization, which introduces a dependency between these two components.

CRediT authorship contribution statement

Jingqiang Chen: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Yunyun Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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